

CSE 4417 IoT Theory

Lecture 05: Networking

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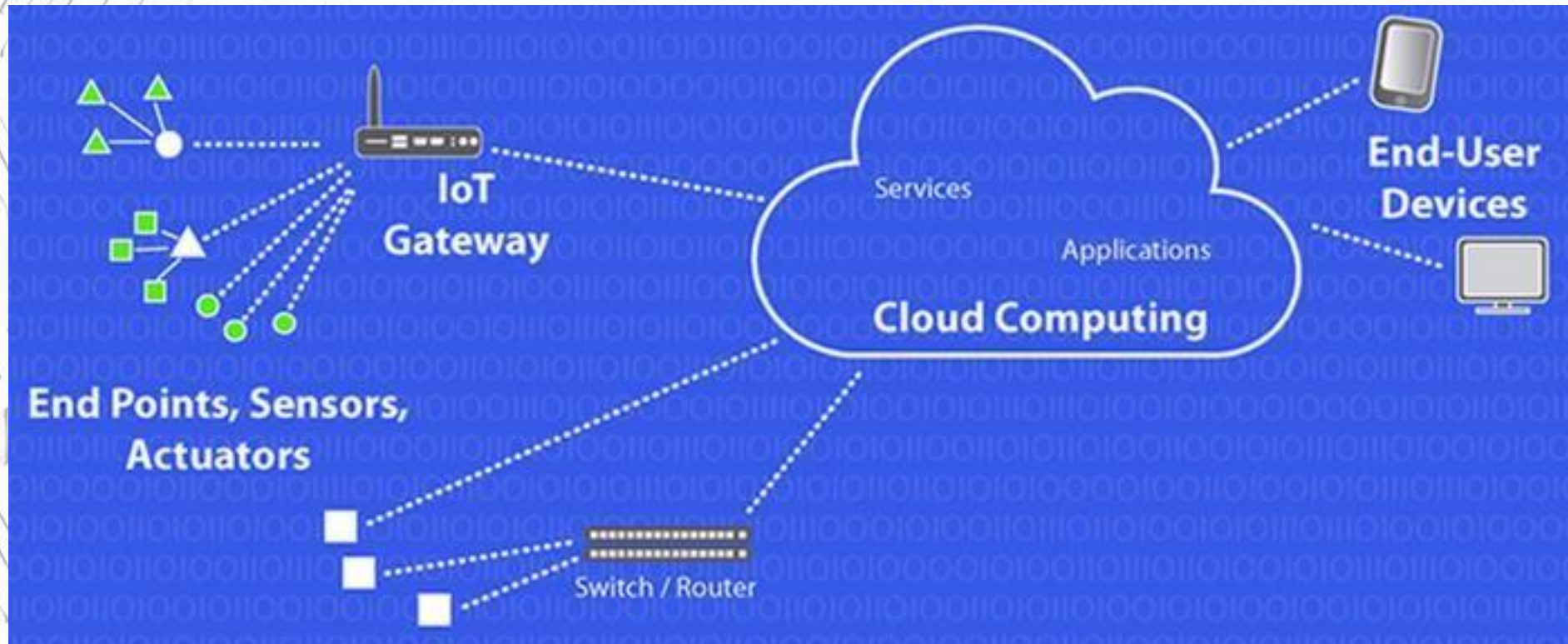
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IoT Components

– From Device to Applications

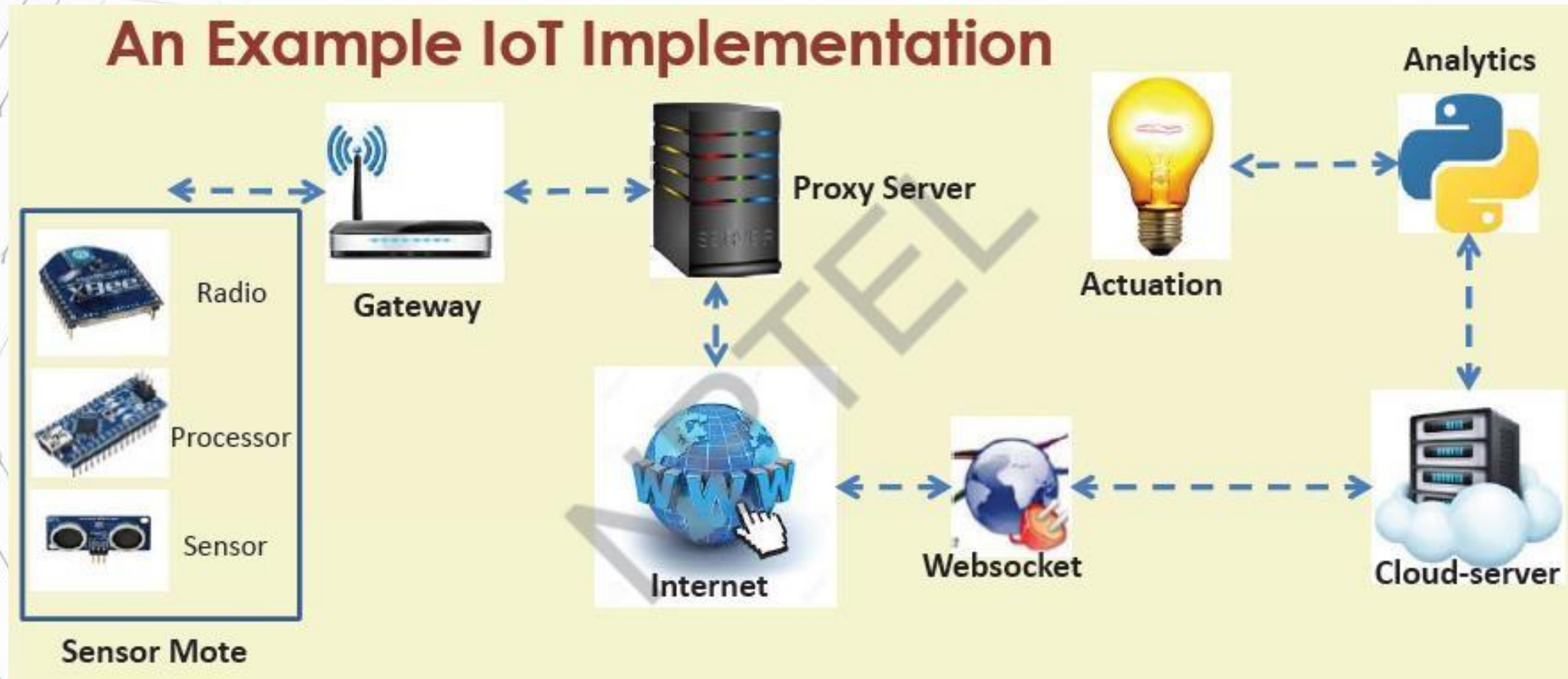


-  **Communication**
-  **Processing**
-  **Analytics**
-  **Server/ Storage**

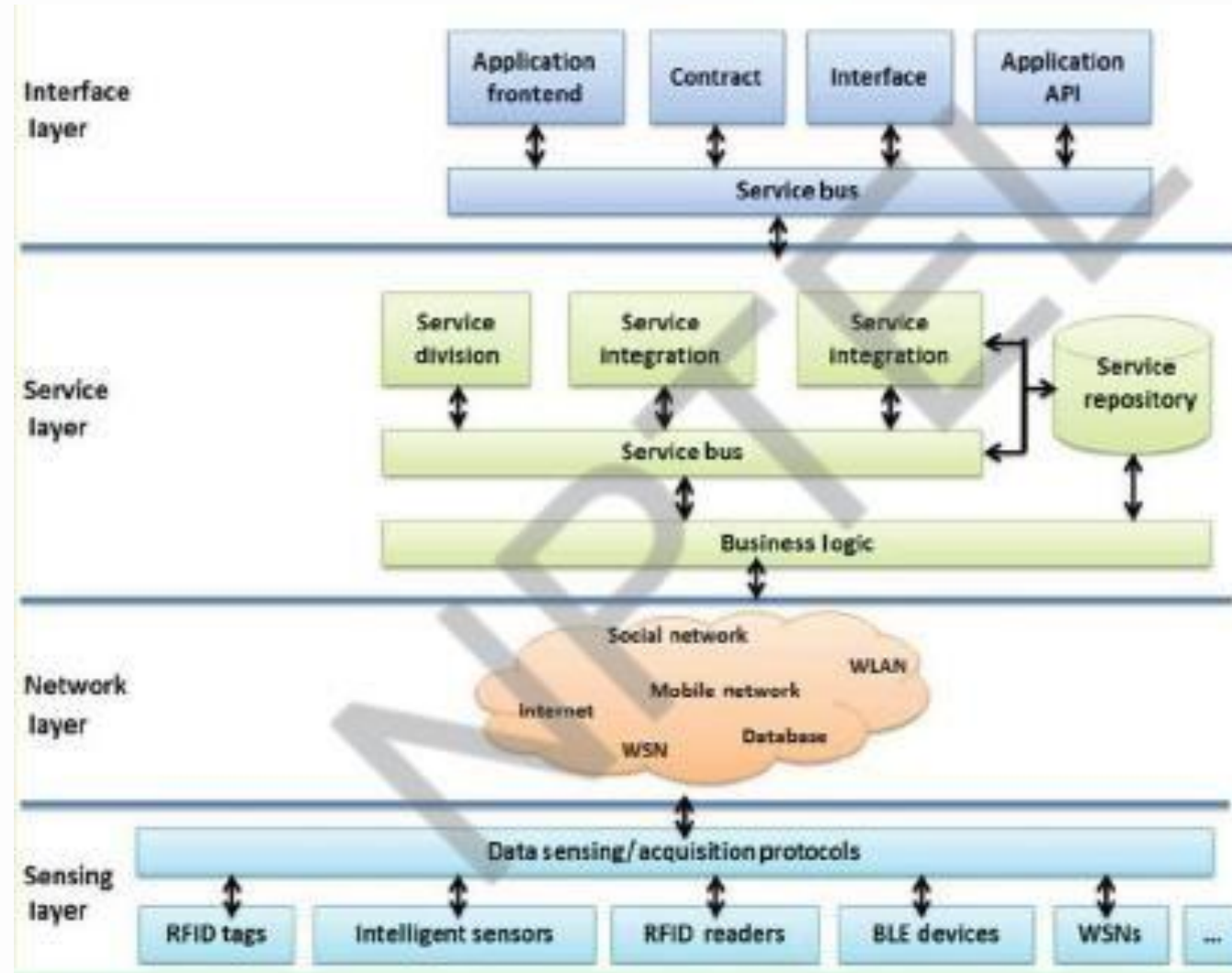
Functional Components of IoT

- Component for interaction and communication with other IoT devices
- Component for processing and analysis of operations
- Component for Internet interaction
- Components for handling Web services of applications
- Component to integrate application services
- User interface to access IoT

Example IoT Implementation



IoT Service Oriented Architecture



IoT Categories

Industrial IoT

- IoT device connects to an IP network and the global Internet
- Communication between the nodes done using regular as well as industry specific technologies.

Consumer IoT

- IoT device communicates within the locally networked devices.
- Local communication is done mainly via Bluetooth, Zigbee or, WiFi.
- Generally limited to local communication by a Gateway

IoT Challenges

- Security
- Scalability
- Energy efficiency
- Bandwidth management
- Modeling and Analysis
- Interfacing
- Interoperability
- Data storage
- Data Analytics
- Complexity management (e.g., SDN)

Complexity of Networks

- Growth of networks
- Interference among devices
- Network management
- Heterogeneity in networks
- Protocol standardization within network

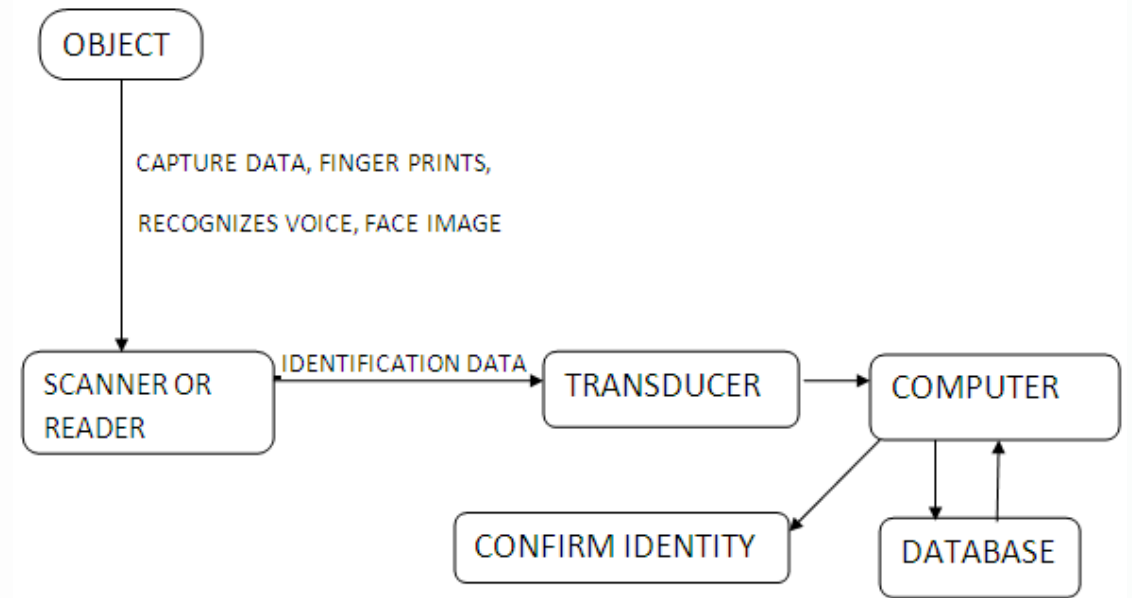
Scalability

- Flexibility within Internet
- IoT integration
- Large scale deployment
- Real-time connectivity of billions of device

Connectivity Technologies

RFID

- RFID is an acronym for “radio-frequency identification”
- Data digitally encoded in RFID tags, which can be read by a reader.
- Somewhat similar to barcodes.
- Data read from tags are stored in a database by the reader.
- As compared to traditional barcodes and QR codes, RFID tag data can be read outside the line-of-sight



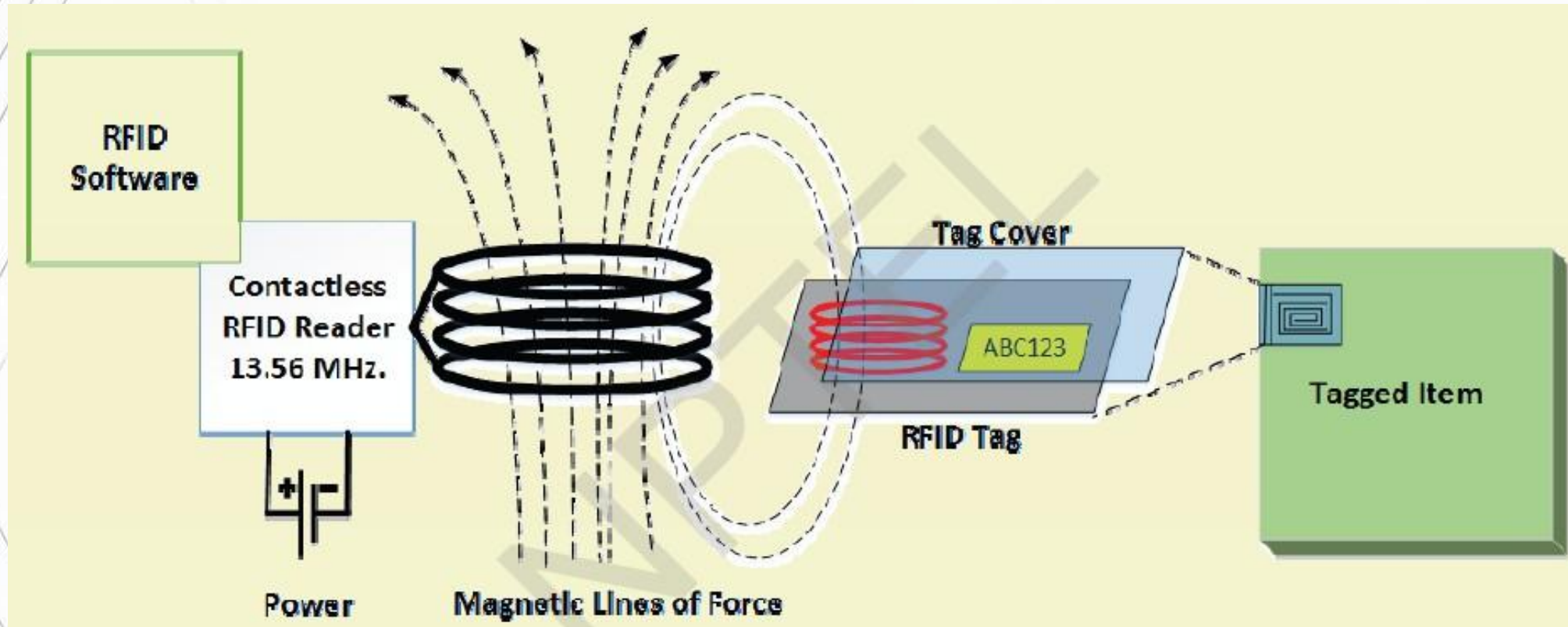
RFID Features

- RFID tag consists of an integrated circuit and an antenna.
- The tag is covered by a protective material which also acts as a shield against various environmental effects.
- Tags may be passive or active.
- Passive RFID tags are the most widely used.
- Passive tags have to be powered by a reader inductively before they can transmit information, whereas active tags have their own power supply

RFID - Working Principle

- Derived from Automatic Identification and Data Capture (AIDC) technology.
- AIDC performs object identification, object data collection and mapping of the collected data to computer systems with little or no human intervention. AIDC uses wired communication
- RFID uses radio waves to perform AIDC functions.
- The main components of an RFID system include an RFID tag or smart label, an RFID reader, and an antenna

RFID



RFID - Applications

- Inventory management
- Asset tracking
- Personnel tracking
- Controlling access to restricted areas
- ID badging
- Supply chain management
- Counterfeit prevention (e.g. in the pharmaceutical industry)



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Thank You

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